

EDUCATION

Charlottesville, VA	University of Virginia	Aug 2016 - May 2020
• Major: Computer Science, B.A. Minors: Economics, Astronomy		
New York, NY	Baruch College (Pre-MFE Coursework)	Jan 2026 -
• Advanced Calculus for Financial Applications (Distinction), Probability Theory for Financial Applications		

EMPLOYMENT

Twilio - Senior Software Engineer	July 2022 - Present
• Building Twilio's next-generation AI-native communication platform, enabling AI agents to participate in multi-channel business interactions with persistent conversational state and AI-to-AI routing. • As a member of TaskRouter, scaled from 50M to 120M+ monthly tasks while maintaining 99.999% uptime, transforming it into one of Twilio's most reliable and high-traffic systems powering the Flex platform, including critical emergency and healthcare services. • Designed and implemented a robust rate-limiting solution adopted across Twilio's core library, now widely used by multiple teams and regularly consulted on implementation two years post-launch. • Part of a leadership-initiated task force addressing a severe outage in the Conversations platform; identified and resolved Sev0 bugs, implemented key enhancements, and helped restore customer confidence. Now driving backend modernization to scale from thousands to millions of MAUs. • Lead high-pressure on-call rotations overseeing millions of real-time customer interactions, including emergency and healthcare services. Diagnosed issues under pressure and implemented post-incident improvements to drive long-term system resilience. • Promoted to Senior Engineer at age 25, one of the youngest among 1,500+ engineers across Twilio.	
Goldman Sachs - Software/Quant Dev	May 2021 – July 2022
• As a member of the Inventory Management team, built trading and inventory-management logic for Goldman Sachs' repo desk, systemizing validation and auto-execution of thousands of daily security movements (billions in firm inventory) directly impacting desk liquidity and P&L. • Promoted within 8 months and entrusted to design and implement Flow Type Auto Execution Thresholds — a new control mechanism improving the safety and efficiency of repo trade validation. • Integrated Delphi with a new downstream API to connect to a new source of U.S. Treasuries held by the firm, increasing available liquidity and boosting desk P&L. • Served in high-pressure 24-hour on-call rotations supporting sub-24-hour repo allocations, resolving time-sensitive issues that directly affected Goldman's revenue.	
Fannie Mae - Software Engineer	June 2020 - May 2021
• Managed sensitive data to build complex financial reports used throughout all levels of the business.	

RESEARCH & PROJECTS

Macro Economic Research and Projects: https://adsoccer1.github.io/
• Series 3 – National Commodities Futures Examination, FINRA — <i>Passed August 2024</i> • Developed Python-based macro models and historical backtests analyzing policy transmission, yield-curve dynamics, FX positioning, and cross-asset liquidity regimes.

Skills

- **Finance & Markets:** Macro trading, STIR (SOFR), U.S. yield curve, FX strategy, systematic strategies
- **Software Engineering:** Java, Python, SQL, AWS, distributed systems, real-time messaging, Kafka, C++, Git, Javascript, React
- **AI Systems:** MCP server implementation, multi-agent orchestration, context management & memory, event-driven workflow